

TRACE

Video Demo Script

2–5 minute screen recording with voice-over · target ≈ 3.5 min

A judge should understand the whole project from this video alone: the problem, the solution, the features, and a live demo. Below, **SAY** is what you narrate; **SHOW** is what's on screen. Before recording: start the backend (:8000) and frontend (:5173), hard-refresh the browser, and have the role-switcher ready.

The Script

0:00–0:20 Hook

SAY Every year, exam-paper leaks destroy the careers of millions of honest students. One insider with early access can compromise an entire exam — and once it's out, nobody can prove whose copy it was. This is Trace: a paper impossible to open before the exam, and traceable to its exact source if it ever leaks.

SHOW Title screen, or the Trace dashboard header ("Exam Security Ops").

0:20–0:45 The problem

SAY Two things go wrong in real life. First, the insider leak — someone between the printer and the exam hall opens the paper early. Second, the accountability gap — even when a leak is caught, there's no way to trace it back, so no one is ever held responsible.

SHOW A simple problem slide, or stay on the dashboard.

0:45–1:15 The solution

SAY Trace solves both with real cryptography. Three guarantees: the paper is encrypted with AES-256 and the key is split across five custodians with Shamir's Secret Sharing — any three needed — plus a server-enforced time lock. Every copy carries an invisible watermark that

survives a screen photo. And every action is written to a tamper-evident audit log. Plus two layers that crush the leak itself: dynamic per-candidate papers and a leak-match detector.

SHOW Slowly scroll the four role chips — Admin, Custodian, Candidate, Investigator.

1:15–1:45 Seal a paper • Admin

SAY As the exam controller I seal a new exam. I choose Dynamic mode — instead of one paper, the system assembles a unique paper for each candidate from an encrypted question bank. I set the release to thirty seconds and seal it.

SHOW Admin → Seal a New Exam → Dynamic bank toggle → blueprint → +30s → Seal & Distribute. Point to the vault, the time gate, and the Dynamic Assembly strip.

1:45–2:15 Unlock ceremony • Custodians

SAY Now the custodians. Each holds one key share, masked until the release window. As they submit, the quorum fills — two of three, three of three. But even with every valid share in, the vault stays sealed until the clock hits zero. That's the time gate. And it opens.

SHOW Custodian → submit a share → switch custodian → submit. Back to Admin ops panel → the vault animates UNLOCKED.

2:15–2:45 Unique watermarked papers • Candidate

SAY Here's the candidate's own paper, stamped with an invisible watermark unique to them. Now switch to a different candidate — the questions are different. Every candidate gets a content-distinct, individually-fingerprinted paper, so a leak exposes only one variant and points straight at its owner.

SHOW Candidate cand001 → watermarked paper + fingerprint. Role-switch to cand002 → visibly different questions.

2:45–3:25 Catch the leak • Investigator

SAY Now the leak. Suppose the paper turns up on a chat group. I paste the text into the Leak-Match Detector — it matches the questions against the bank and names the suspect, R-001. If instead I have a photo, Forensic Trace recovers the watermark even after JPEG re-compression and identifies the source at 100% confidence. Every detection becomes a case file with the candidate's full details, and the whole investigation is sealed in a SHA-256 hash chain — Verify Integrity: intact.

SHOW Investigator → Leak-Match (paste text) → suspect. Forensic Trace (upload image) → traced. Case Files → open the case → full candidate card. Verify Integrity → chain intact.

3:25–3:40 Close

SAY Trace makes early opening cryptographically impossible, every copy traceable, and every record tamper-proof — backed by 86 passing tests and crypto built from scratch. Leaks become impossible to hide, and impossible to deny. Thank you.

SHOW *Return to the dashboard header; optional end card with the three guarantees.*

Before You Hit Record

- Backend running: `cd backend && uvicorn trace.api.app:app → http://127.0.0.1:8000`
- Frontend running: `cd frontend && npm run dev → http://localhost:5173` (hard-refresh once)
- Pre-seed a dynamic exam that is already UNLOCKED so you don't wait on the timer live — or set release to +20s and edit the pause out.
- Have one candidate's paper image downloaded and its text copied, ready to use as the leak.

Demo Logins

Role	Username	Password
Admin / Exam Controller	admin	admin123
Custodians (1-5)	cust1 ... cust5	custodian1 ... custodian5
Candidate	cand001 / cand002	candidate1 / candidate2
Investigator	investigator	invest123

If a judge asks: "Is the crypto real?"

Yes — AES-256-GCM via pycryptodome, but Shamir's Secret Sharing is implemented from scratch over the $GF(256)$ finite field with its own unit tests; the watermark is a Koch-Zhao DCT scheme; the audit log is a real SHA-256 hash chain. 86 tests prove it, including that no two custodians can ever reconstruct the key.

Trace — built for India's FAR AWAY 2026. Real cryptography, real tracing, real audit.